

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID2026-2287
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	3 MARKS

3.3 – Data Flow Diagram

1. Ingestion Phase (The Input)

- **Data Sources:** Raw real estate metrics are pulled simultaneously from **Zillow** (for transaction listings, geocoded locations, and historical sales) and **Kaggle** (for property characteristics like square footage, age, and room counts).
- **Geospatial Layering:** Supplementary third-party data, such as census demographics and neighborhood walkability scores, are introduced to add localized context.

2. Processing & Transformation Phase (The Pipeline)

- **Data Cleaning (ETL):** The raw streams route through an automated cleaning pipeline designed to handle missing attributes, standardize mismatched formatting, and validate geocodes for accurate mapping.
- **Integration & Blending:** Once sanitized, the independent datasets undergo a **Geospatial Join** and schema integration, merging the physical house features with their exact geographical coordinates and market timing windows.

3. Analytics & Calculation Phase (The Logic)

- **Feature Engineering:** The system calculates real-time variables, mapping out **Price per Sq Ft**, assessing variable impact ratios (**Feature Importance**), and generating market velocity metrics (**Average Days on Market**).
- **KPI Synthesis:** Aggregated data feeds the core analytics engines to track multi-year trends, neighborhood price hot spots, and market trend vectors.

4. Visualization Phase (The Output)

- **Dashboard Layering:** The final, processed stream branches into targeted Tableau endpoints:
 - An **Interactive Map Layer** displaying localized pricing distribution.

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- A **Feature Contribution Tree Map / Scatterplot** showing which variables alter property values the most.

